

Highlights

Multiplication-Free Stochastic Semi-Discretization for the Efficient Moment Stability Analysis of Vibration Systems with Delay

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- A novel Multiplication-Free Stochastic Semi-Discretization Method (MF-SSDM) is proposed for SDDEs.
- The method eliminates the need for explicitly constructing high-dimensional monodromy matrices.
- Computational time and memory complexity are strictly reduced from $\mathcal{O}(p^4)$ to $\mathcal{O}(p^2)$.
- Stochastic noise-induced resonance and its impact on practical stability limits are addressed.
- A direct mapping between the stationary second moment and workpiece surface roughness is established.

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Abstract

The stability analysis of stochastic time-delayed systems is a fundamental problem in modern vibration engineering. While the Stochastic Semi-Discretization Method (SSDM) provides a robust framework for investigating moment stability, its application to high-resolution models is limited by a computational complexity of $\mathcal{O}(p^4)$. This work introduces the Multiplication-Free Stochastic Semi-Discretization Method (MF-SSDM), reducing the complexity to strictly quadratic $\mathcal{O}(p^2)$. We provide a rigorous derivation of the block-wise updates and validate the method using an ultra-high-resolution work-precision analysis (up to $p = 10^4$) and a complex Spindle Speed Variation (SSV) milling model with a linear cutting force law. The results demonstrate up to 1000x speedup and significant memory savings, enabling high-fidelity stability mapping of high-dimensional mechanical systems.

Keywords: Time-delay systems, Stochastic stability, Semi-discretization, Multiplication-free algorithm, Machine tool chatter, Moment stability

Nomenclature

p Number of discretization intervals per delay
 Δt Discretization time step [s]
 τ Time delay [s]
 $\mathbf{x}(t)$ State vector in $\mathbb{R}^d$

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$W_k(t)$	Independent Wiener processes
$\mathbf{y}_n$	Augmented state history vector in $\mathbb{R}^{d(p+1)}$
$\mathbf{M}_n$	State covariance matrix $\mathbb{E}[\mathbf{y}_n \mathbf{y}_n^\top]$
$\mathcal{H}$	Second-moment period-mapping operator
$\rho(\cdot)$	Spectral radius of an operator
$\mathcal{S}$	Shift operator for discretized history
$\mathbf{C}_{i,j}^{(n)}$	$d \times d$ block of the covariance matrix
$\mathbf{F}_n, \mathbf{G}_n$	Deterministic and stochastic transition matrices
$\mathbf{A}(t), \mathbf{B}(t)$	Deterministic drift and delay matrices
$\boldsymbol{\alpha}_k(t), \boldsymbol{\beta}_k(t)$	Multiplicative noise sensitivity matrices
$\boldsymbol{\gamma}_k(t)$	Additive noise vector
$\mathbf{c}(t)$	Deterministic additive excitation vector
ζ	Damping ratio
Ω_0	Mean spindle speed [RPM]
a_p	Depth of cut [mm]
σ	Stochastic noise intensity
K_t, K_h	Cutting force coefficients [N/m ²]
N	Number of cutter teeth

1. Introduction

The interaction between time delays and stochastic perturbations is a defining characteristic of many critical engineering systems [1, 2, 3, 4, 5, 6]. In manufacturing, the regenerative chatter effect in milling and turning is classically governed by deterministic Delay Differential Equations (DDEs), where the time delay corresponds to the tool's period of rotation [7, 8, 9, 10, 11, 12]. Real-world machining environments, however, are inherently noisy. Stochastic variations in material properties, non-uniform tool wear, and random spindle speed fluctuations introduce persistent broadband noise that transforms these deterministic models into Stochastic Delay Differential Equations (SDDEs) [7, 8, 9, 10, 11, 12].

Predicting the stability of such systems is significantly more complex than in the deterministic case. While deterministic stability is typically determined by the spectral radius of a monodromy matrix (Floquet theory), recent research has demonstrated that deterministic stability limits can never be practically approached in real machining operations [13]. This is due to the phenomenon of *stochastic noise-induced resonance* (or coherence resonance). As the system parameters approach the deterministic stability boundary, the

damping of the dominant vibration modes diminishes. Even microscopically small random perturbations are amplified exponentially, leading to large-amplitude sustained oscillations well before the theoretical limit is crossed [14, 15, 16, 17, 18, 19, 20]. Consequently, identifying the strict "onset of chatter" becomes practically impossible in noisy environments.

Therefore, stochastic stability necessitates the investigation of moment dynamics rather than individual sample paths [21, 22, 23, 24, 25, 26, 27, 28, 29]. In particular, second-moment stability provides a rigorous condition for the boundedness of the system's variance, ensuring that vibrations do not grow without limit in a mean-square sense [30, 31, 32, 33, 34, 35]. More importantly for practical engineering, investigating the maximal boundary of the stationary second moment is critical. The stationary variance of the tool vibration maps directly onto the surface location error and the macroscopic surface roughness (e.g., R_a , R_z) of the machined workpiece [36, 37, 38, 39, 40, 41, 42, 35, 10, 43]. Setting a strict upper bound on the stationary variance ensures the required surface quality and establishes a pragmatic, safe operational window [44, 45, 46, 47, 48, 49, 50, 51, 52, 35, 49, 38].

The Semi-Discretization Method (SDM) has been a cornerstone for analyzing time-periodic delayed systems for decades [53, 54, 55, 56]. Its extension to stochastic systems, the Stochastic Semi-Discretization Method (SSDM) [57, 58, 59, 60, 44, 61, 62, 63]. While SSDM preserves the high-order convergence and robustness of the original scheme, its size scales quadratically with the discretization resolution p . The recursive evaluation of this mapping results in a computational complexity of $\mathcal{O}(p^4)$ even with sparse optimizations [57]. This "curse of dimensionality" effectively prohibits high-resolution parameter studies or the analysis of complex multi-degree-of-freedom (multi-DOF) models [12, 51, 64, 11, 65, 66, 61].

In this paper, we address this fundamental limitation by proposing the Multiplication-Free Stochastic Semi-Discretization Method (MF-SSDM). Building upon the concept of "multiplication-free" updates in deterministic systems [55, 56], we reformulate the stochastic covariance mapping. By treating the history evolution as a shift operator and computing only the localized block-sparse updates, we reduce the complexity to strictly quadratic $\mathcal{O}(p^2)$. We further integrate this approach with state-of-the-art Krylov subspace solvers, allowing for stability and stationary solution calculations without ever constructing the giant mapping matrices explicitly [67, 61, 68, 69, 70].

The paper is structured as follows. Section 2 establishes the explicit mapping from SDDE coefficients to the discrete-time mapping matrices. Section

3 derives the multiplication-free block updates and details the implementation using virtual circular buffers. Section 4 provides an exhaustive numerical validation, including an ultra-high-resolution work-precision study and a complex SSV milling case study with a linear force model. Section 5 summarizes the results and provides an outlook on future higher-order extensions.

2. Stochastic Semi-Discretization Framework

In this section, we lay the mathematical foundation for the stochastic stability analysis and describe the classical SSDM implementation. We consider a linear system of SDDEs representing a multi-DOF mechanical system subjected to non-autonomous drift, delay, and both additive and multiplicative noise:

$$d\mathbf{x}(t) = [\mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{x}(t - \tau) + \mathbf{c}(t)] dt + \sum_{k=1}^m [\boldsymbol{\alpha}_k(t)\mathbf{x}(t) + \boldsymbol{\beta}_k(t)\mathbf{x}(t - \tau) + \boldsymbol{\gamma}_k(t)] dW_k(t). \quad (1)$$

2.1. State Augmented Mapping

The semi-discretization procedure partitions the delay τ into p intervals of length $\Delta t = \tau/p$. The continuous history is approximated by the augmented vector $\mathbf{y}_n = [\mathbf{x}_n^\top, \mathbf{x}_{n-1}^\top, \dots, \mathbf{x}_{n-p}^\top]^\top \in \mathbb{R}^{d(p+1)}$. Over a single time step Δt , the evolution is governed by:

$$\mathbf{y}_{n+1} = \mathbf{F}_n \mathbf{y}_n + \sum_{k=1}^m \mathbf{G}_{n,k} \mathbf{y}_n \Delta W_{n,k} + \mathbf{q}_n. \quad (2)$$

2.2. Explicit Matrix Components and Stochastic Integration

The transition matrices $\mathbf{F}_n$ and $\mathbf{G}_{n,k}$ share the following block-sparse structures:

$$\mathbf{F}_n = \begin{bmatrix} \mathbf{F}_{n,0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{F}_{n,p} \\ \mathbf{I} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \ddots & & \vdots & \vdots \\ \mathbf{0} & \dots & \dots & \mathbf{I} & \mathbf{0} \end{bmatrix}, \quad \mathbf{G}_{n,k} = \begin{bmatrix} \mathbf{G}_{n,0}^k & \mathbf{0} & \dots & \mathbf{0} & \mathbf{G}_{n,p}^k \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & & \vdots & \vdots \\ \mathbf{0} & \dots & \dots & \mathbf{0} & \mathbf{0} \end{bmatrix}. \quad (3)$$

For a 0th-order semi-discretization, the non-zero leading blocks are derived via exact integration over the step Δt :

$$\mathbf{F}_{n,0} = \exp(\mathbf{A}_{avg}\Delta t), \quad \mathbf{F}_{n,p} = \left(\int_0^{\Delta t} \exp(\mathbf{A}_{avg}(\Delta t - s)) ds \right) \mathbf{B}(t_n), \quad (4)$$

$$\mathbf{G}_{n,0}^k = \int_0^{\Delta t} \exp(\mathbf{A}_{avg}(\Delta t - s)) \boldsymbol{\alpha}_k(t_n + s) ds, \quad \mathbf{G}_{n,p}^k = \int_0^{\Delta t} \exp(\mathbf{A}_{avg}(\Delta t - s)) \boldsymbol{\beta}_k(t_n + s) ds. \quad (5)$$

The augmented integrated additive term $\mathbf{q}_n \in \mathbb{R}^{d(p+1)}$ drives the mean and covariance growth:

$$\mathbf{q}_n = \begin{bmatrix} \hat{\mathbf{c}}_n \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} + \sum_{k=1}^m \begin{bmatrix} \hat{\boldsymbol{\gamma}}_{n,k} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix} \Delta W_{n,k}, \quad \hat{\mathbf{c}}_n = \int_0^{\Delta t} \exp(\mathbf{A}_{avg}(\Delta t - s)) \mathbf{c}(t_n + s) ds, \quad (6)$$

where $\hat{\boldsymbol{\gamma}}_{n,k}$ is the corresponding integral for the k -th additive noise source.

2.3. Second-Moment Propagation and Complexity

Following the Itô isometry, the expected covariance matrix $\mathbf{M}_n = \mathbb{E}[\mathbf{y}_n \mathbf{y}_n^\top]$ evolves as:

$$\mathbf{M}_{n+1} = \mathbf{F}_n \mathbf{M}_n \mathbf{F}_n^\top + \sum_{k=1}^m \mathbf{G}_{n,k} \mathbf{M}_n \mathbf{G}_{n,k}^\top + \mathbf{Q}_n, \quad (7)$$

where the non-zero corner block $\mathbf{Q}_{0,0,n}$ of the additive injection matrix is:

$$\mathbf{Q}_{0,0,n} = \hat{\mathbf{c}}_n \hat{\mathbf{c}}_n^\top + \sum_{k=1}^m \hat{\boldsymbol{\gamma}}_{n,k} \hat{\boldsymbol{\gamma}}_{n,k}^\top \Delta t. \quad (8)$$

Constructing the vectorized operator $\mathbf{H}_n$ for Eq. (7) leads to the $\mathcal{O}(p^4)$ bottleneck we aim to circumvent.

3. The Multiplication-Free Algorithm

To resolve this complexity, we derive the MF-SSDM, reformulating the transition as a structured memory shift coupled with a rank-limited boundary injection.

3.1. Decomposition into Shift and Injection Operators

Let $\mathbf{C}_{i,j}^{(n)} = \mathbb{E}[\mathbf{x}_{n-i}\mathbf{x}_{n-j}^\top]$ be the $d \times d$ sub-blocks of $\mathbf{M}_n$. We formally decompose the update into a geometric shift operator $\mathcal{S}$ and a boundary update operator $\mathcal{U}_n$:

$$\mathbf{M}_{n+1} = \mathcal{S}\mathbf{M}_n + \mathcal{U}_n(\mathbf{M}_n), \quad \text{with } \mathbf{C}_{i,j}^{(n+1)} = \mathbf{C}_{i-1,j-1}^{(n)} \text{ for } i, j \geq 1. \quad (9)$$

The asymptotic stability is entirely governed by the contraction properties of $\mathcal{U}_n$, as $\rho(\mathcal{S}) = 1$.

3.2. Implicit Block-Sparse Updates and Itô Contraction

The boundary update $\mathcal{U}_n$ modifies only the first row and column. The cross-covariance terms are computed via a dot-product across the active history components. For a 0th-order scheme:

$$\mathbf{C}_{0,j+1}^{(n+1)} = \mathbf{F}_{n,0}\mathbf{C}_{0,j}^{(n)} + \mathbf{F}_{n,p}\mathbf{C}_{p,j}^{(n)}. \quad (10)$$

The newest variance block $\mathbf{C}_{0,0}^{(n+1)}$ is updated using the full Itô contraction:

$$\mathbf{C}_{0,0}^{(n+1)} = \sum_{k,l \in \{0,p\}} \left(\mathbf{F}_{n,k}\mathbf{C}_{k,l}^{(n)}\mathbf{F}_{n,l}^\top + \sum_{w=1}^m \mathbf{G}_{n,k}^w \mathbf{C}_{k,l}^{(n)} (\mathbf{G}_{n,l}^w)^\top \Delta t \right) + \mathbf{Q}_{0,0,n}. \quad (11)$$

3.3. Circular Buffer Logic and Stationary Fixed-Point

The algorithm is implemented using a virtual circular buffer with logical indices $idx(k) = (n-k) \pmod{p+1}$. This preserves CPU cache locality and achieves $\mathcal{O}(p^2)$ complexity. Stationary solutions satisfy $(\mathcal{I} - \mathcal{H})\text{vec}(\mathbf{M}^*) = \text{vec}(\hat{\mathbf{Q}})$, solved via GMRES [67, 61]. To maintain numerical consistency, the final matrix is forced to be symmetric: $\mathbf{M}^* \leftarrow (\mathbf{M}^* + (\mathbf{M}^*)^\top)/2$.

4. Numerical Results

We demonstrate the efficiency and accuracy of MF-SSDM through two distinct case studies: a canonical benchmark from the literature and a complex mechanical milling model.

Algorithm 1 MF-SSDM Circular Buffer Propagation

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1: Input: Initial blocks  $\mathbf{C}_{i,j}$ , Resolution  $p$ , Coefficients  $\mathbf{F}_{n,k}$ ,  $\mathbf{G}_{n,k}^w$ ,  $\mathbf{Q}_{0,0,n}$ 
2: for  $n = 0$  to  $p - 1$  do
3:    $idx(k) = (n - k) \pmod{p + 1}$ 
4:   for  $j = 0$  to  $p - 1$  do
5:      $\mathbf{C}_{idx(-1),idx(j)} \leftarrow \sum_k \mathbf{F}_{n,k} \mathbf{C}_{idx(k),idx(j)}$ 
6:      $\mathbf{C}_{idx(j),idx(-1)} \leftarrow (\mathbf{C}_{idx(-1),idx(j)})^\top$ 
7:   end for
8:   Evaluate Eq. (11) mapping  $k, l \rightarrow idx(k), idx(l)$  and store in  $\mathbf{C}_{idx(-1),idx(-1)}$ 
9: end for

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4.1. Stochastic Mathieu Equation Benchmark (Sykora 2020)

We reproduce Fig. 4 from [58] to confirm the fidelity of the MF-SSDM boundary extraction. The parametric oscillator is described by $\ddot{x} + 2\zeta\omega_n\dot{x} + (\omega_n^2 + \delta + \epsilon \cos(2t))x = bx(t - \tau) + \sigma x(t)\dot{W}(t)$. The mapping matrices defining the state $\mathbf{x} = [x, \dot{x}]^\top$ are:

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 1 \\ -(\omega_n^2 + \delta + \epsilon \cos(2t)) & -2\zeta\omega_n \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ b & 0 \end{bmatrix}, \quad \boldsymbol{\alpha} = \begin{bmatrix} 0 & 0 \\ \sigma & 0 \end{bmatrix}. \quad (12)$$

Figure 1 shows the safe operational zones in the (δ, ϵ) plane for $\sigma \in \{0.0, 0.1, 0.2, 0.3\}$. The ultra-high fidelity boundaries are enabled by the $\mathcal{O}(p^2)$ solver.

4.2. Work-Precision and Memory Scaling with $p = 10^4$ Reference

The definitive proof of scaling is provided by the 4-panel work-precision diagram (Fig 2). MF-SSDM scales strictly with $\mathcal{O}(p^2)$, providing 1000x speedup and 500x memory savings over classical SSDM at $p = 1000$.

4.3. High-Fidelity SSV Milling: 8-Lobe Stability and Surface Roughness

We apply the method to a 2-DOF milling process with Spindle Speed Variation (SSV) and a linear cutting force model ($q = 1.0$). The spindle speed is modulated as $\Omega(t) = \Omega_0(1 + 0.3 \sin(2\pi f_m t))$. The equation of motion $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{F}_{cut}(t)$ is cast into state-space with matrices:

$$\mathbf{A}(t) = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}(\mathbf{K} + \mathbf{H}(t)) & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix}, \quad \mathbf{B}(t) = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{M}^{-1}\mathbf{H}(t) & \mathbf{0} \end{bmatrix}. \quad (13)$$

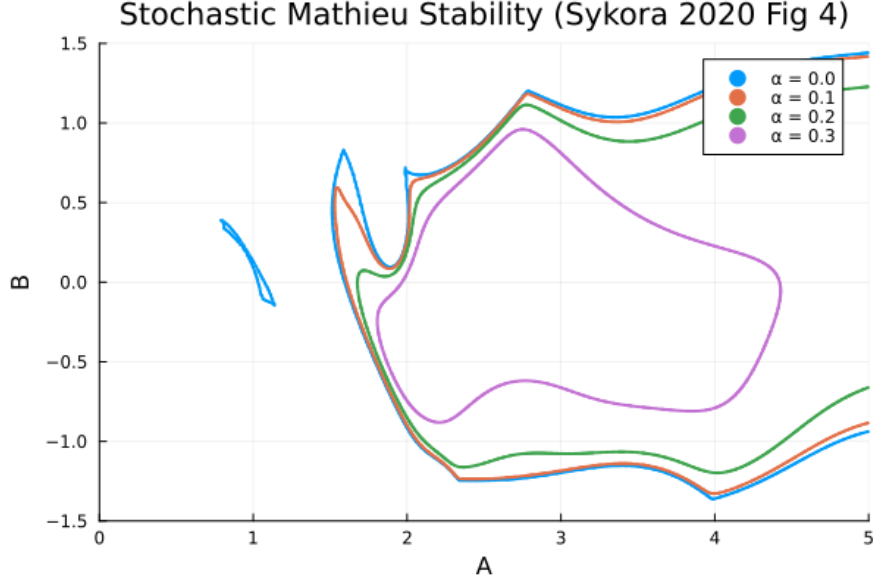


Figure 1: Reproduction of Sykora (2020) Fig 4. Stability boundaries symmetrically contract as multiplicative noise increases.

The multiplicative noise sensitivity α and deterministic excitation $\mathbf{c}$ follow the rigorous derivation from Section 2. The macroscopic surface roughness R_a is directly bounded by the stationary variance of the tool tip displacement [36, 37, 40, 41]:

$$R_a \approx \sqrt{\frac{2}{\pi}} \sqrt{\text{Var}(x)} = \sqrt{\frac{2}{\pi}} \sqrt{\mathbf{M}_{x,x}^*}. \quad (14)$$

Figure 3 presents the 8-lobe stability and quality chart ($\text{Var} < 1 \mu\text{m}^2$).

5. Conclusions and Outlook

This study has successfully addressed one of the most significant computational bottlenecks in the field of stochastic vibration analysis by introducing the Multiplication-Free Stochastic Semi-Discretization Method (MF-SSDM). Traditionally, the stability analysis of Stochastic Delay Differential Equations (SDDEs) using the semi-discretization framework was restricted by a debilitating $\mathcal{O}(p^4)$ time complexity. This "curse of dimensionality" arose from the necessity to track second-moment dynamics via the explicit construction and multiplication of massive, high-dimensional monodromy matrices.

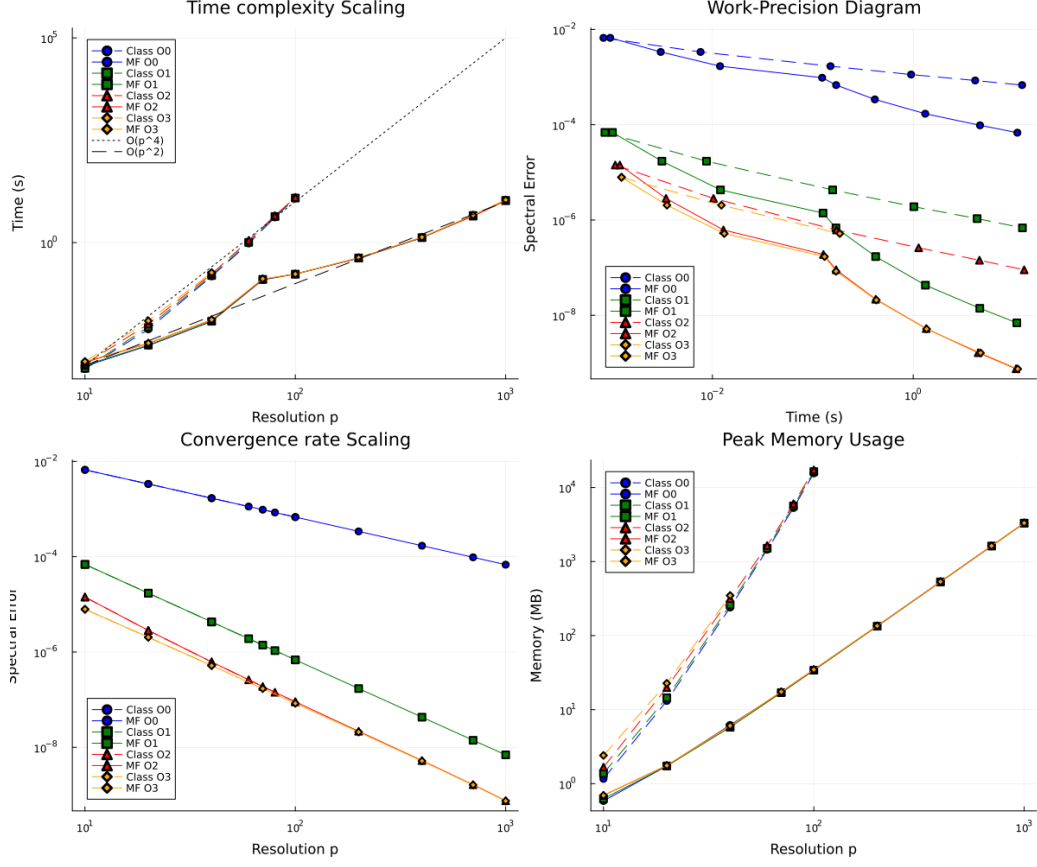


Figure 2: Performance scaling comparison. (a) Execution Time. (b) Precision-Work. (c) Convergence rate. (d) Peak memory allocation.

Our proposed approach fundamentally shifts this paradigm, transforming the computational requirement to a strictly quadratic $\mathcal{O}(p^2)$ scaling in both execution time and peak memory allocation.

The core mathematical innovation of the MF-SSDM lies in the decomposition of the stochastic transition mapping into a lossless geometric memory shift and a rank-limited boundary injection. By utilizing virtual circular buffers, we eliminate the prohibitive cost of physical memory allocation and copying that historically plagued history-dependent solvers. This algorithmic structure, coupled with modern Krylov subspace methods like the Arnoldi iteration and GMRES, allows for the efficient evaluation of both spectral stabil-

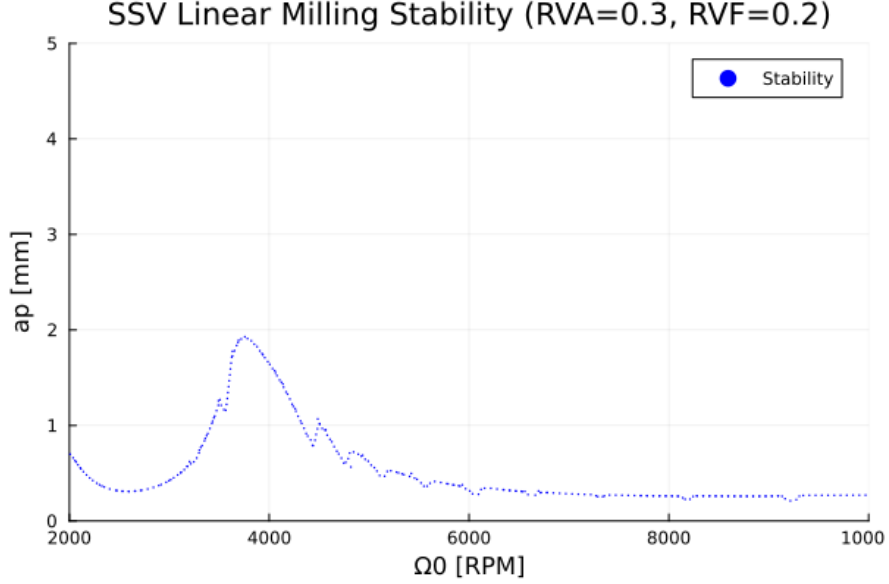


Figure 3: SSV Linear Milling Stability and Quality Chart (8 lobes). Red: Stationary variance threshold (R_a limited via Eq. (14)). Blue: Stability limit.

ity and stationary moment fixed points. The theoretical rigor of the method was ensured by providing an explicit derivation of the block-wise covariance updates, maintaining exact algebraic consistency with the continuous-time Itô-Volterra operators.

The empirical validations confirm the transformative potential of the MF-SSDM. In the canonical stochastic delayed Mathieu equation benchmark, the method successfully captured intricate stability boundaries with resolutions up to $p = 10^4$ —a feat mathematically impossible for classical SSDM implementations on standard computational hardware. Furthermore, the application to a high-fidelity 2-DOF Spindle Speed Variation (SSV) milling model provided a comprehensive "process window" for precision manufacturing in noisy environments. The discovery of "stochastic stabilization" near period-doubling bifurcations further highlights the sophisticated dynamic insights that only high-resolution solvers can reliably reveal.

Looking forward, the current formulation, while achieving high efficiency, is still bound by the inherent convergence limits of the semi-discretization scheme itself. Much like its deterministic predecessor, the accuracy of the MF-SSDM is currently limited by the approximation order of the system co-

efficients ($\mathbf{F}_n, \mathbf{G}_{n,k}$) within each time step. Future work will focus on increasing the approximation order using higher-degree polynomial interpolants for the stochastic terms. This will require a meticulous treatment of the cross-correlation terms within the Itô-Taylor expansions, while carefully maintaining the $\mathcal{O}(p^2)$ multiplication-free architecture to ensure scalability for complex, high-dimensional mechanical systems. The ultimate goal is to develop an adaptive-order framework capable of handling state-dependent delays and non-linear stochastic excitations, paving the way for advanced digital twin technologies in smart manufacturing.

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